FAB FLOW / SEMICONDUCTOR MATERIAL HANDLING RESEARCH

Deadlock-Aware OHT Routing and Partial Rail Reservation

반도체 FAB의 OHT 경로·교차로 예약·강화학습을 통합한 합성 시뮬레이션 연구

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01 RESEARCH QUESTION

최단경로와 전체 처리량은 다르다. 차량 한 대가 가장 빠른 길을 찾더라도, 여러 차량이 같은 교차로를 공유하면 시스템 전체의 물류는 지연될 수 있다.

Full-route reservation

출발 시 전체 경로의 레일 자원을 잠가 교착을 보수적으로 회피하지만, 미사용 레일까지 선점한다.

Partial next-segment reservation

다음 진입 구간만 검사·예약하고 통과한 구간은 해제해 레일 활용률을 높인다.

Wait-for Graph

차량 A가 B를 기다리는 관계를 추적해 순환대기 위험이 커지는 진입을 사전에 거부한다.

Question — Does learning itself improve throughput once the same safety constraints and routing structure are held constant?

02 SIMULATION MODEL

Virtual FAB topology · 48 nodes, 82 directed rail links, 9 OHT vehicles.

3 scenarios × 5 policies × 40 paired seeds = 600 evaluation runs; 360 safety/fleet sensitivity runs.

Arrival model	Synthetic Poisson transport requests
Normal demand	68 requests / hour
Demand surge	1.5× normal arrivals
Observation	1,500 s (25 minutes)
Learning	Tabular Q-learning
Statistics	Bootstrap CIs / Holm-corrected sign test

03 STATISTICAL EVIDENCE

DEMAND SURGE / COMPLETED JOBS



Partial dynamic vs full-route: +13.425 completed jobs (95% paired CI [+11.650, +15.125]).

Trained Q vs untrained Q: +0.575 jobs (95% CI [+0.200, +1.000]); Holm-adjusted p = 0.0369.

The learned effect was not consistently significant under normal and rail-closure scenarios.

Main finding: Resource reservation strategy mattered more than additional Q-learning in this synthetic test. Not a real factory digital twin or industrial safety-certified controller.